

# Monthly Airbnb Demand Forecasting – Chicago

Course: IT 5413 – Forecasting

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## Abstract

This project aims to forecast monthly Airbnb demand in Chicago using time series forecasting techniques. The objective is to analyze historical Airbnb review data, identify underlying patterns, and develop models that can accurately predict future demand. The dataset was aggregated into monthly counts and transformed using a logarithmic function to stabilize variance and improve model performance.

Exploratory data analysis revealed a clear seasonal pattern along with a gradual upward trend in Airbnb demand. Based on these observations, multiple forecasting models were developed, including baseline models (Mean, Naïve, Seasonal Naïve, and Drift), Exponential Smoothing (ETS), Auto ARIMA, STL decomposition combined with ETS, and a Neural Network Autoregression model. The dataset was divided into training and testing sets, with the last six months reserved for evaluating model performance.

Model accuracy was assessed using key performance metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). Among all models, the Auto ARIMA model demonstrated the best performance, achieving the lowest RMSE of 0.0386, MAE of 0.0279, and MAPE of 0.304. The STL+ETS model produced comparable results with an RMSE of 0.0392, while the ETS model showed significantly higher error with an RMSE of 0.118. The neural network model performed comparatively poorly, with an RMSE of approximately 0.205, indicating that it was less suitable for this dataset.

A combination model was also tested by averaging forecasts from ARIMA, ETS, and STL+ETS models. However, the combined model resulted in a higher RMSE of 0.0556, suggesting that including weaker models reduced overall forecasting accuracy. Based on these findings, the ARIMA model was selected as the final model.

The final model was re-fitted using the entire dataset and used to generate a 12-month forecast. The results indicate stable seasonal demand patterns in Airbnb activity, providing useful insights for planning and decision-making in the hospitality industry.

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## **1. Introduction**

Time series forecasting is a statistical technique used to analyze historical data and predict future values over time. It is widely applied in various fields such as finance, economics, healthcare, and business analytics. By identifying patterns such as trends, seasonality, and cyclical behavior, time series models help organizations make informed decisions based on expected future outcomes. In today's data-driven environment, accurate forecasting plays a crucial role in planning, resource allocation, and strategic decision-making.

In the hospitality and tourism industry, forecasting demand is particularly important due to the presence of strong seasonal fluctuations. Demand for accommodation often varies depending on factors such as holidays, weather conditions, local events, and travel trends. Platforms like Airbnb have significantly transformed the accommodation market, providing flexible and dynamic options for travelers. As a result, understanding and predicting Airbnb demand has become essential for hosts, property managers, and businesses to optimize pricing strategies, improve occupancy rates, and manage resources efficiently.

The primary objective of this project is to forecast monthly Airbnb demand in Chicago using time series forecasting techniques. This involves analyzing historical Airbnb review data, identifying key patterns, and applying multiple forecasting models to determine the most accurate approach. The study compares several models, including baseline methods, Exponential Smoothing (ETS), Auto ARIMA, decomposition-based models, and neural network approaches. The goal is to evaluate their performance using statistical accuracy measures and select the best model for future forecasting.

This problem is important because accurate demand forecasting can provide valuable insights for decision-making in the short-term rental market. For example, better forecasts can help hosts adjust pricing during peak seasons, prepare for periods of high demand, and avoid losses during low-demand periods. Additionally, understanding demand patterns can support city-level planning and tourism management. By developing a reliable forecasting model, this project aims to contribute to more effective and data-driven decision-making in the Airbnb ecosystem.

## **2. Literature Review**

Time series forecasting is widely used across fields such as finance, economics, and tourism to predict future values based on historical data. The primary objective is to identify patterns such as trends, seasonality, and randomness, which are essential for accurate forecasting and decision-making.

Simple benchmark models, such as the Naïve and Seasonal Naïve methods, are commonly used as baseline approaches due to their simplicity (Hyndman & Athanasopoulos, 2021). However, these models often fail to capture underlying patterns in the data, limiting their effectiveness for more complex time series.

More advanced methods include Exponential Smoothing (ETS), which models level, trend, and seasonality by assigning greater weight to recent observations (Gardner, 2006). ARIMA models, introduced by Box and Jenkins (1970), are among the most widely used forecasting techniques, as they combine autoregressive and moving average components while applying differencing to achieve stationarity. Seasonal ARIMA models further extend this approach to capture repeating seasonal patterns.

Decomposition-based methods such as STL (Cleveland et al., 1990) separate a time series into trend, seasonal, and residual components, allowing for more accurate modeling of seasonal effects. Additionally, machine learning approaches such as neural networks have been applied to capture nonlinear relationships (Zhang, 2003). However, their effectiveness depends on the availability of large datasets and complex patterns.

Forecast combination techniques aim to improve accuracy by averaging predictions from multiple models, although their success depends on the quality of individual models (Hyndman & Athanasopoulos, 2021).

Overall, existing research suggests that no single model consistently outperforms others. Model performance depends on the characteristics of the dataset. This study contributes by comparing multiple models on Airbnb demand data and demonstrating that simpler models, such as ARIMA, can outperform more complex approaches when the data structure is relatively straightforward.

### **3. Methodology**

This study applies a structured time series forecasting approach to predict monthly Airbnb demand in Chicago. The methodology includes data preparation, exploratory analysis, model development, and model evaluation.

#### **3.1 Data Description and Preparation**

The dataset consists of Airbnb review data for Chicago, used as a proxy for demand. Daily records were aggregated into monthly counts to create a consistent time series, reducing noise and highlighting long-term patterns such as trend and seasonality.

The data was converted into a time-indexed format, and only recent years (post-2022) were retained to ensure stability. A logarithmic transformation ( $\log_{1p}$ ) was applied to stabilize variance and reduce the impact of extreme values.

### 3.2 Exploratory Data Analysis

Exploratory analysis revealed a clear upward trend and repeating seasonal patterns. STL decomposition was used to separate the series into trend, seasonal, and remainder components. The results confirmed strong seasonality, indicating the need for models capable of capturing seasonal behavior.

### 3.3 Train-Test Split

The dataset was divided into training and testing sets, with the last six months reserved for testing. This approach allows for objective evaluation of model performance on unseen data. A six-month forecast horizon was used for consistency across models.

### 3.4 Forecasting Models

Several models were implemented to capture different aspects of the data:

- **Baseline Models:** Mean, Naïve, Seasonal Naïve, and Drift models were used as reference points.
- **ETS Model:** Captures level, trend, and seasonality using weighted observations.
- **ARIMA Model:** Automatically selects optimal parameters and incorporates seasonal differencing ( $D = 1$ ) to capture repeating patterns.
- **STL + ETS Model:** Combines decomposition with ETS to improve seasonal modeling.
- **Neural Network (NNETAR):** Applied to capture potential nonlinear relationships.
- **Combination Model:** Averages forecasts from multiple models to reduce model-specific errors.

### 3.5 Model Evaluation

Models were evaluated using RMSE, MAE, and MAPE. The model with the lowest RMSE was selected as the best-performing model. The final model was then re-fitted on the full dataset to generate a 12-month forecast.

### 3.6 Justification of Approach

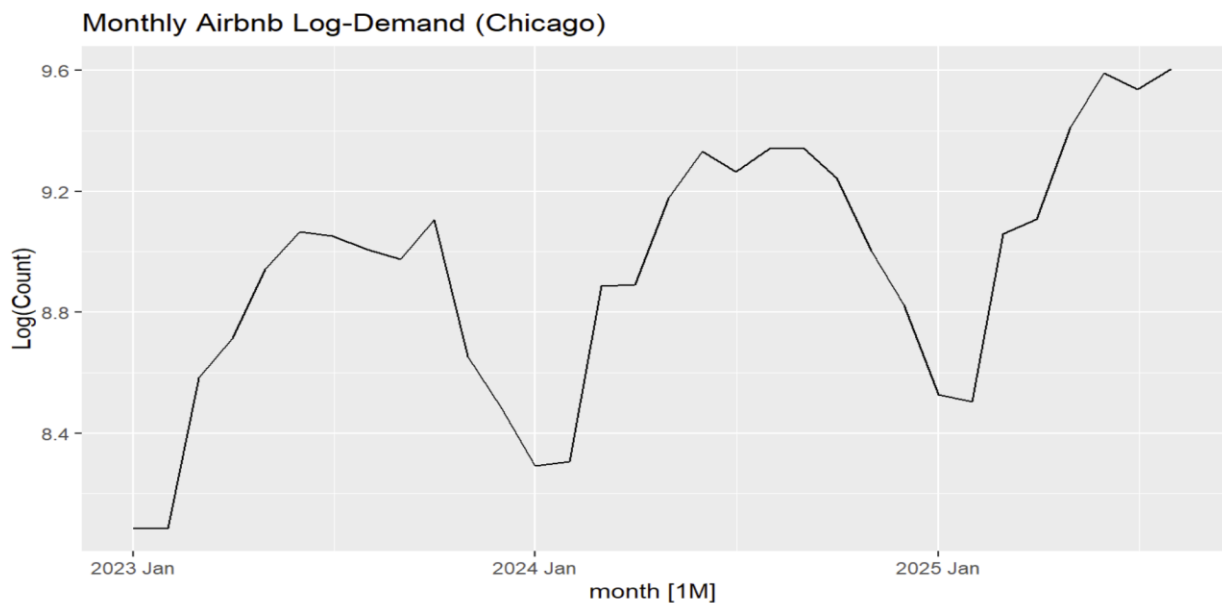
The methodology is appropriate given the strong seasonal patterns in the data. ARIMA and STL-based models are well-suited for such structures, while baseline and neural network models provide useful comparisons. The train-test split ensures objective evaluation, and the combination model tests the robustness of forecasts.

## 4. Results

This section presents the findings from the time series forecasting models applied to the Airbnb demand dataset. The results include visualizations, model accuracy comparisons, and interpretation of forecasting performance.

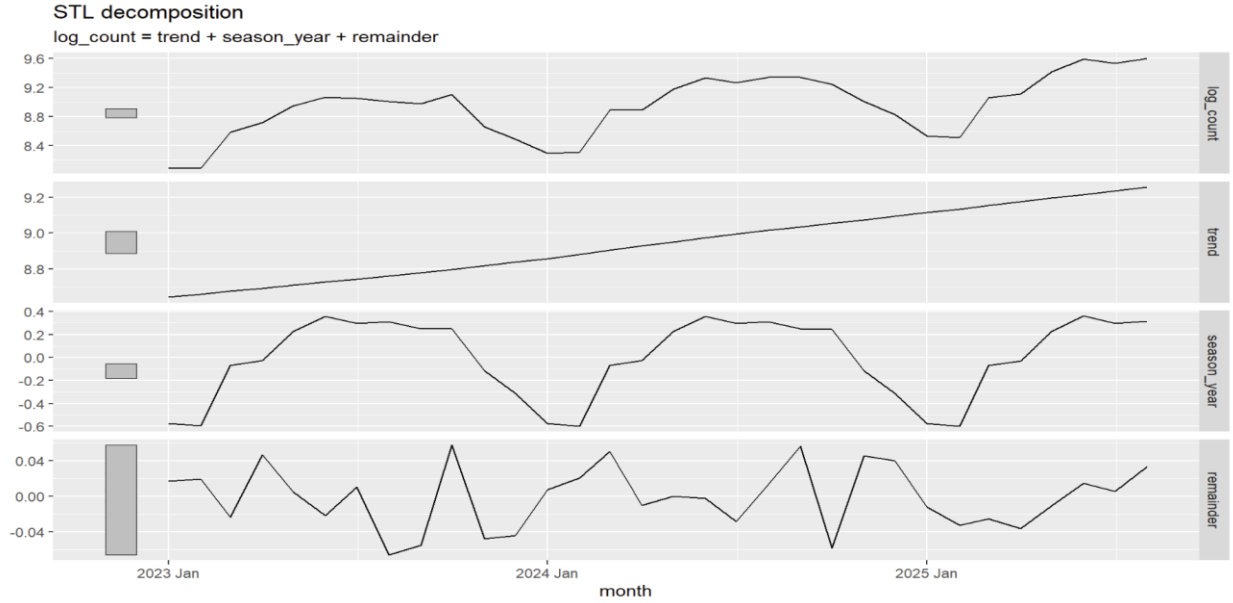
### 4.1 Exploratory Analysis Results

The time series plot of monthly Airbnb demand (Figure 1) shows a clear upward trend along with recurring seasonal patterns. This indicates that demand increases over time while also following a repeating yearly cycle.



**Figure 1: Monthly Airbnb Demand in Chicago (Log-Transformed)**

The STL decomposition results (Figure 2) further confirm that the data consists of three key components: trend, seasonal, and remainder. The seasonal component shows consistent fluctuations across months, while the remainder component remains relatively small. This suggests that most of the variation in the data can be explained by trend and seasonality.



**Figure 2: STL Decomposition of Monthly Airbnb Demand (Trend, Seasonality, and Remainder)**

## 4.2 Baseline Model Performance

The baseline models were evaluated first to establish a reference point for comparison. The results show that these models performed poorly in terms of forecasting accuracy.

| Model          | RMSE  | MAE   | MAPE |
|----------------|-------|-------|------|
| Naïve          | 0.909 | 0.881 | 9.34 |
| Drift          | 0.846 | 0.823 | 8.72 |
| Mean           | 0.594 | 0.551 | 5.82 |
| Seasonal Naïve | 0.239 | 0.236 | 2.51 |

**Table 1: Baseline Model Performance Comparison**

Among the baseline models, the Seasonal Naïve model performed relatively better, as it accounts for repeating seasonal patterns. However, its accuracy remained significantly lower than that of more advanced models.

## 4.3 Advanced Model Performance

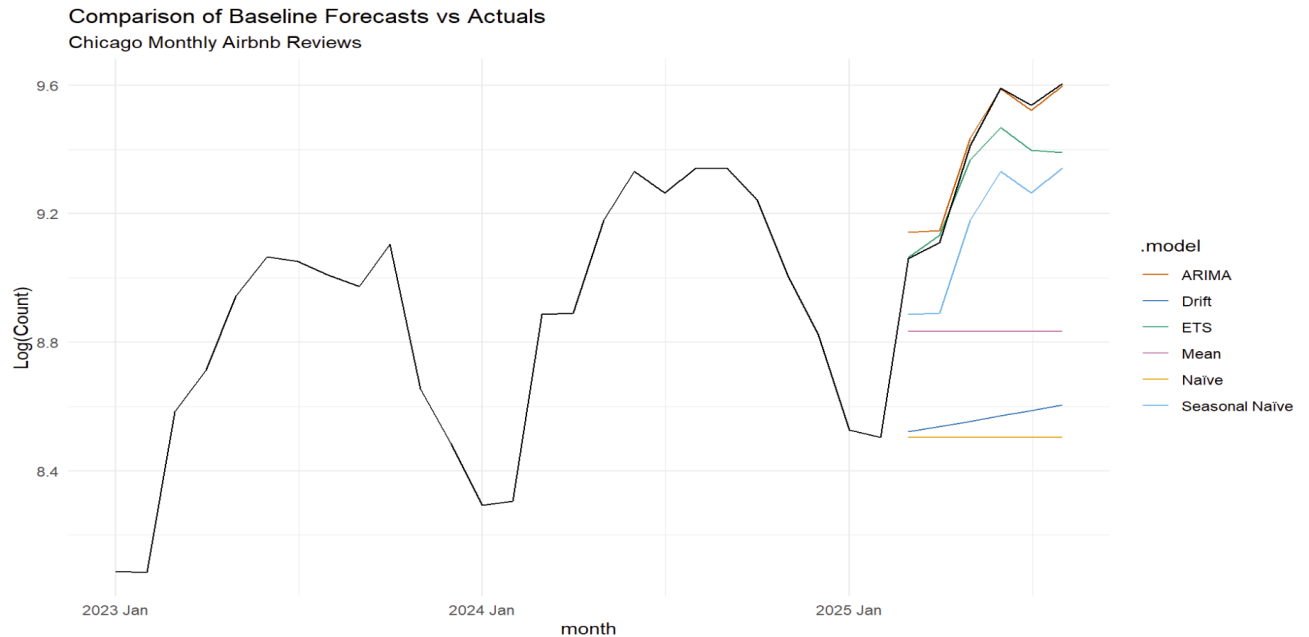
More advanced forecasting models were evaluated to improve forecasting accuracy beyond baseline approaches. The results are summarized in Table 2.

| Model     | RMSE   | MAE    | MAPE  |
|-----------|--------|--------|-------|
| ARIMA     | 0.0386 | 0.0279 | 0.304 |
| STL + ETS | 0.0392 | 0.0331 | 0.350 |
| ETS       | 0.118  | 0.0913 | 0.957 |
| NNETAR    | 0.205  | 0.172  | 1.81  |
| Combo     | 0.0556 | 0.0482 | 0.509 |

**Table 2: Advanced Model Performance Comparison**

Among the advanced models, the ARIMA model performed best, achieving the lowest RMSE value of 0.0386. The STL+ETS model also produced comparable results, indicating that decomposition-based methods effectively capture seasonal patterns. The ETS and neural network models showed lower accuracy, suggesting that they are less suitable for this dataset. Although combination models are often expected to improve performance, in this case the combined model performed worse than ARIMA alone. This is because the inclusion of weaker models reduced the overall accuracy of the combined forecast. The significant reduction in RMSE from baseline models (0.239) to advanced models (0.0386) highlights the effectiveness of advanced forecasting techniques.

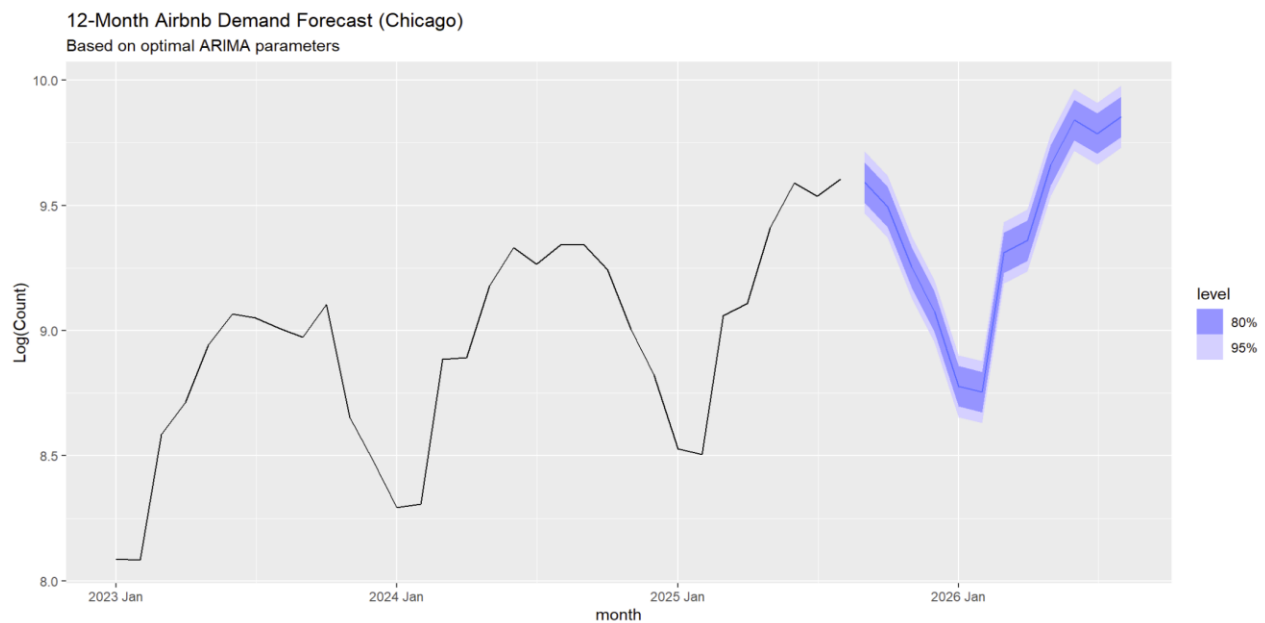
Figure 3 presents the forecast comparison across models, showing that the ARIMA model closely follows the observed values, while other models exhibit larger deviations.



**Figure 3: Forecast Comparison Across Models**

## 4.4 Final Forecast Results

After selecting the best-performing model, the ARIMA model was re-fitted using the entire dataset. A 12-month forecast was generated, as shown in Figure 4.



**Figure 4: Final 12-Month Forecast**

The figure shows the 12-month forecast generated using the ARIMA model. The black line represents historical data, while the blue line shows forecasted values with 80% and 95% confidence intervals. The forecast indicates a continued seasonal pattern, with demand expected to decrease initially and then increase in later months. The relatively narrow confidence intervals suggest that the model provides reliable predictions.

## 4.5 Significance of Findings

The results highlight the importance of selecting an appropriate model based on data characteristics. The ARIMA model performed best as it effectively captured the seasonal structure of the dataset. Simpler models failed to account for these patterns, while more complex models such as neural networks did not improve performance due to limited data and lack of strong nonlinear behavior.

These findings demonstrate that increased model complexity does not necessarily lead to better accuracy. Instead, models that align with the underlying structure of the data produce more reliable forecasts.

## 5. Discussion

The results indicate that forecasting performance depends heavily on how well the model aligns with the structure of the data. The Airbnb dataset exhibits clear seasonal patterns with relatively simple structure, which explains why the ARIMA model performed best. The STL+ETS model also produced similar results, confirming the importance of capturing seasonality.

Baseline models performed poorly as they failed to account for trend and seasonal components. Although the ETS model improved upon baseline approaches, it was still less accurate than ARIMA. The neural network model did not perform well, likely due to limited data size and lack of strong nonlinear relationships.

These findings are consistent with existing literature, which suggests that ARIMA models remain highly effective for time series data with clear seasonal patterns. Similarly, decomposition-based methods such as STL are known to perform well in such cases. The results also support the idea that more complex models do not always lead to better performance.

One limitation of this study is the relatively small dataset, which may restrict the effectiveness of advanced models such as neural networks. Additionally, the analysis does not include external factors such as economic conditions, weather, or events that may influence Airbnb demand.

Future work could include incorporating additional variables, extending the dataset, and exploring hybrid models to improve forecasting accuracy.

## 6. Conclusion

This study applied multiple time series forecasting models to predict Airbnb demand in Chicago. The ARIMA model achieved the best performance, effectively capturing seasonal patterns and producing the most accurate forecasts.

The results demonstrate that selecting a model based on data characteristics is more important than using more complex methods. The final forecast indicates stable seasonal demand patterns, providing useful insights into planning and decision-making.

Future research can focus on incorporating external variables and expanding the dataset to further enhance forecasting performance.

## 7. References

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